

UNIT 1.7 - Properties of determinants

In this short unit we discuss those properties of determinants which enable us to compute or expand a determinant even easier than before.

You will immediately see that these properties remind one of the method for row reduction of matrices. It is important to note the difference between these two processes. (Notes, Theorem 14 p 36.)

By exchanging two rows of a matrix one obtains a row equivalent matrix, but when exchanging two rows of a determinant, one actually changes the sign of the initial determinant.

Furthermore the multiplication of a row of a determinant by a real number does change the value of this determinant.

Also note that the determinant of rA is not equal to r times the determinant of A . Actually the value of the determinant of rA depends on the size of A !

An interesting feature of a determinant is, that whatever you may do with the rows, you may also do with the columns of your determinant. That means the theorems remain valid if the word **ROW** is substituted with the word **COLUMN**. (Notes, Theorems 14 and 15 pp 36-37.)

We use these theorems to "simplify" a determinant before calculating it.

Consider the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ -4 & 5 & 6 \\ 7 & -8 & 9 \end{bmatrix}$.

As we know it is easier to evaluate a determinant if there are zeros in a row/column and therefore I will use the theorems to get as many zeros as possible in a row/column. I will use the "1" in the first row and first column to do so:

$$\det(A) = \begin{vmatrix} 1 & 2 & 3 \\ -4 & 5 & 6 \\ 7 & -8 & 9 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 \\ 0 & 13 & 18 \\ 0 & -22 & -12 \end{vmatrix} \begin{matrix} \\ R_2 \rightarrow R_2 + 4R_1 \\ R_3 \rightarrow R_3 - 7R_1 \end{matrix}.$$

Now expand using the first column:

$$\det(A) = \begin{vmatrix} 1 & 2 & 3 \\ 0 & 13 & 18 \\ 0 & -22 & -12 \end{vmatrix} = (-1)^{1+1} 1 \begin{vmatrix} 13 & 18 \\ -22 & -12 \end{vmatrix} + 0 + 0 = -156 + 396 = 240$$

If I want to expand the determinant using the first row:

$$\det(A) = \begin{vmatrix} 1 & 2 & 3 \\ -4 & 5 & 6 \\ 7 & -8 & 9 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ -4 & 13 & 18 \\ 7 & -22 & -12 \end{vmatrix} = 240.$$

Sometimes, I can use the theorems to find the value of a determinant without expanding.

$$\text{Let } \det(A) = \begin{vmatrix} a & b & c \\ 1 & 2 & 3 \\ d & e & f \end{vmatrix} = k$$

Then

$$\text{(i)} \quad \begin{vmatrix} a & b & c \\ d & e & f \\ 1 & 2 & 3 \end{vmatrix} = - \begin{vmatrix} a & b & c \\ 1 & 2 & 3 \\ d & e & f \end{vmatrix} = -k \quad (\text{Theorem 14(b)})$$

$$\begin{aligned} \text{(ii)} \quad \begin{vmatrix} 1 & 2 & 3 \\ 3a & 3b & 3c \\ -2d & -2e & -2f \end{vmatrix} &= 3 \begin{vmatrix} 1 & 2 & 3 \\ a & b & c \\ -2d & -2e & -2f \end{vmatrix} = 3 \times (-2) \begin{vmatrix} 1 & 2 & 3 \\ a & b & c \\ d & e & f \end{vmatrix} \quad (\text{Theorem 14(c)}) \\ &= 3 \times (-2) \times (-1) \begin{vmatrix} a & b & c \\ 1 & 2 & 3 \\ d & e & f \end{vmatrix} = 6k \end{aligned}$$

$$\text{(iii)} \quad |-A| = (-1)^3 |A| = -k \quad (\text{Theorem 16(1)})$$

$$\text{(iv)} \quad \begin{vmatrix} a-1 & b-2 & c-3 \\ d & e & f \\ 1 & 2 & 3 \end{vmatrix} = \begin{vmatrix} a & b & c \\ d & e & f \\ 1 & 2 & 3 \end{vmatrix} \begin{matrix} R_1 - R_3 \\ \\ \end{matrix} \quad (\text{Theorem 14(a)})$$

$$= - \begin{vmatrix} a & b & c \\ 1 & 2 & 3 \\ d & e & f \end{vmatrix} = -k$$